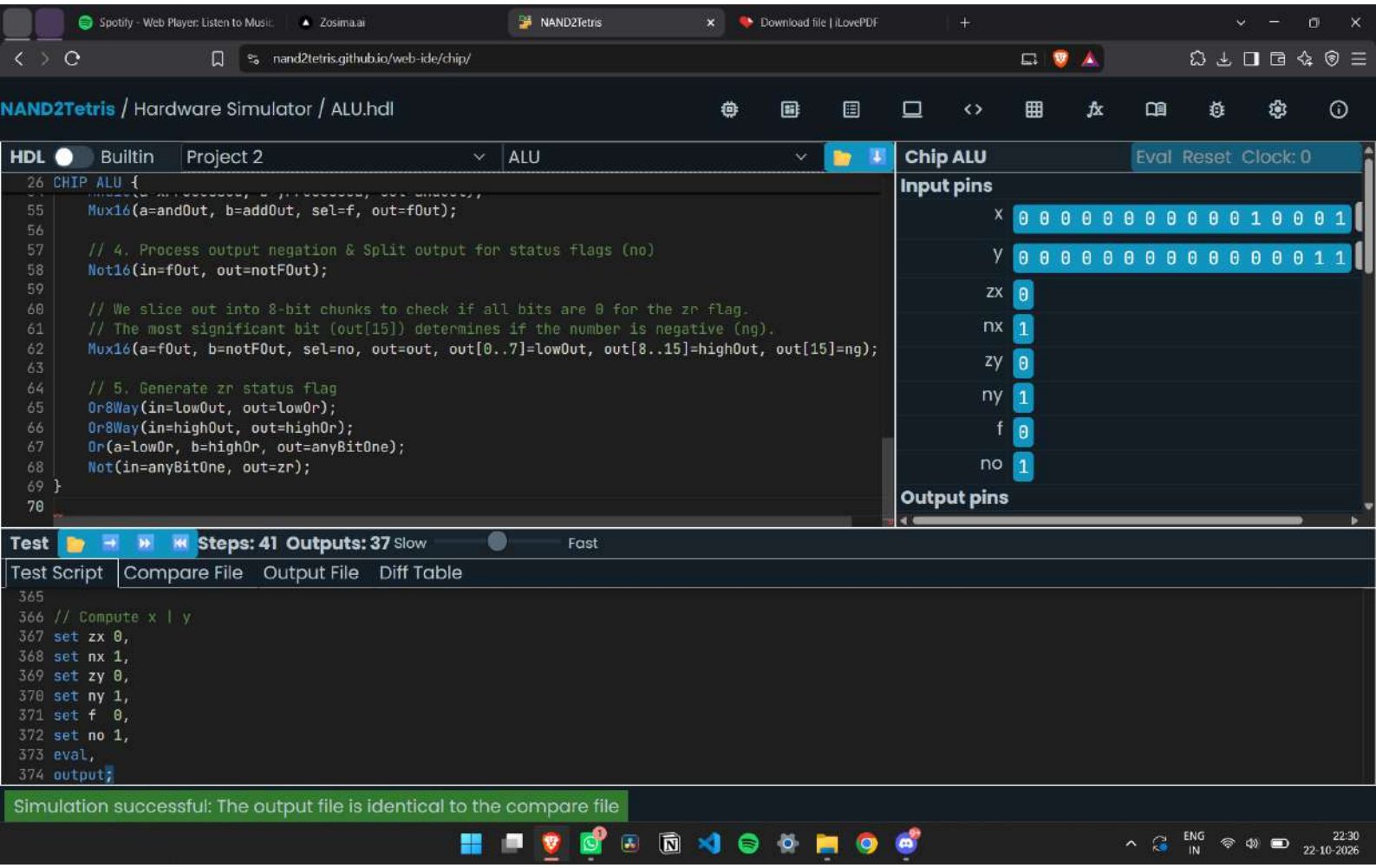

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Zosima.ai
NAND2Tetris
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nand2tetris.github.io/web-ide/chip/
NAND2Tetris / Hardware Simulator / ALU.hdl
HDL
Builtin
Project 2
ALU
CHIP ALU {
Mux16(a=andOut, b=addOut, sel=f, out=fOut);
// 4. Process output negation & Split output for status flags (no)
Not16(in=fOut, out=notFOut);
// We slice out into 8-bit chunks to check if all bits are 0 for the zr flag.
// The most significant bit (out[15]) determines if the number is negative (ng).
Mux16(a=fOut, b=notFOut, sel=no, out=out, out[0..7]=lowOut, out[8..15]=highOut, out[15]=ng);
// 5. Generate zr status flag
Or8Way(in=lowOut, out=lowOr);
Or8Way(in=highOut, out=highOr);
Or(a=lowOr, b=highOr, out=anyBitOne);
Not(in=anyBitOne, out=zr);
}
Chip ALU
Eval Reset Clock: 0
Input pins
x 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 1
y 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1
zx 0
nx 1
zy 0
ny 1
f 0
no 1
Output pins
Test
Steps: 41 Outputs: 37
Slow
Fast
Test Script
Compare File
Output File
Diff Table
// Compute x | y
set zx 0,
set nx 1,
set zy 0,
set ny 1,
set f 0,
set no 1,
eval,
output;
Simulation successful: The output file is identical to the compare file
ENG
IN
22:30
22-10-2026